

Normalization to BCNF – Hotel Management System

1. Introduction

Normalization is the process of organizing data to reduce redundancy and improve data integrity. This document shows normalization up to Boyce-Codd Normal Form (BCNF) for the given system.

2. Initial Relations

Table	Attributes
Guest	guest_id (PK), name, phone, email, address, nationality
Hotel	hotel_id (PK), name, location
RoomType	type_id (PK), name, description
Room	room_id (PK), type_id (FK), hotel_id (FK), room_number, price_per_night, status
Staff	staff_id (PK), hotel_id (FK), name, role
Booking	booking_id (PK), guest_id (FK), check_in_date, check_out_date, total_amount, status
Booking_Room	booking_id (PK, FK), room_id (PK, FK)
Payment	payment_id (PK), booking_id (FK), amount, date, method
Review	review_id (PK), guest_id (FK), rating, comment
Service	service_id (PK), booking_id (FK), type, cost

3. Functional Dependencies

guest_id → name, phone, email, address, nationality

hotel_id → name, location

type_id → name, description

room_id → type_id, hotel_id, room_number, price_per_night, status

staff_id → hotel_id, name, role

booking_id → guest_id, check_in_date, check_out_date, total_amount, status

payment_id → booking_id, amount, date, method

review_id → guest_id, rating, comment

service_id → booking_id, type, cost

4. Normalization Process

First Normal Form (1NF): All tables have atomic values and no repeating groups.

Second Normal Form (2NF): No partial dependency exists since all tables use single-attribute primary keys (except Booking_Room which has a composite key and no partial dependency).

Third Normal Form (3NF): No transitive dependencies are present. Non-key attributes depend only on primary keys.

BCNF: For every functional dependency $X \rightarrow Y$, X is a super key in all tables.

5. BCNF Verification

All determinants in the relations are candidate keys. Therefore, all tables satisfy BCNF conditions.

6. Conclusion

The database schema is fully normalized up to BCNF. No further decomposition is required, ensuring minimal redundancy and maximum consistency.